

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA1507
---------------------	---	---------------------	---------

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Lakshana K [192524433] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Answer

Cloud computing provides **auto-scaling** and **load balancing** to handle sudden increases in website traffic efficiently. During seasonal sales such as Black Friday or festive discounts, thousands of customers may access an e-commerce website at the same time. Without proper resource management, servers may become overloaded, resulting in slow response times or website crashes.

Auto-scaling automatically increases or decreases the number of virtual servers based on user demand. When traffic increases, additional servers are created automatically. During low traffic periods, unnecessary servers are removed, reducing operational costs.

Load balancing distributes incoming user requests evenly across multiple servers. Instead of sending all requests to one server, the load balancer redirects traffic to available servers, preventing overload and improving system performance.

This approach significantly **reduces downtime** because if one server fails, traffic is automatically redirected to healthy servers. Users experience faster page loading, smooth checkout processes, and fewer service interruptions.

Monitoring tools continuously observe CPU usage, memory utilization, response time, and network traffic. If resource usage exceeds predefined limits, monitoring tools trigger auto-scaling and notify administrators about potential issues before they affect users.

Example: During an Amazon Great Indian Festival sale, millions of customers visit the website simultaneously. Auto-scaling launches additional cloud servers, while load balancing distributes customer requests evenly. Monitoring services continuously check system performance to ensure uninterrupted shopping.

Conclusion

Auto-scaling, load balancing, and monitoring together improve performance, reduce downtime, optimize costs, and provide an excellent customer experience during high-traffic events.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Answer

Cloud storage provides a secure and reliable solution for storing backup data. Cloud providers maintain multiple copies of data across different servers and locations, ensuring high durability and protection against hardware failures.

Data redundancy means backup files are stored in multiple data centers. Even if one storage location fails, data remains available from another location. This greatly reduces the risk of permanent data loss.

Cloud providers also offer **automatic backups, snapshots, versioning, and replication**, making data recovery easier after accidental deletion or system failure.

Encryption protects backup data both while it is stored (data at rest) and while it is transferred over the network (data in transit). Even if unauthorized users access the storage, encrypted data cannot be read without the proper encryption key.

Access control is implemented using Identity and Access Management (IAM), Multi-Factor Authentication (MFA), and Role-Based Access Control (RBAC). Only authorized users can view or restore backup data.

Compared to traditional storage systems, cloud storage eliminates the need for purchasing expensive hardware, maintenance, power, cooling, and physical storage space. Organizations only pay for the storage they actually use.

Example: A hospital experiences ransomware that encrypts its local servers. Since backup data is securely stored in cloud storage with replication and versioning enabled, patient records can be restored quickly, ensuring business continuity.

Conclusion

Cloud storage provides secure, durable, cost-effective, and highly available backup solutions while supporting fast disaster recovery.

3. A mobile app company plans to deploy its application globally. Explain how Content Delivery Networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Answer

A **Content Delivery Network (CDN)** is a network of distributed servers that stores copies of application content closer to users. Instead of retrieving data from one central server, users receive content from the nearest CDN server, reducing network delays.

CDNs improve performance by decreasing **latency**, increasing download speed, and reducing the load on the main application server. Frequently accessed files such as images, videos, software updates, and web pages are cached at multiple locations worldwide.

Edge servers are located near users in different countries and regions. When a user requests content, the nearest edge server responds immediately, resulting in faster loading times and a better user experience.

Cloud providers maintain data centers across multiple geographic regions, enabling applications to remain available even if one region experiences technical issues. Traffic is automatically redirected to healthy regions when required.

Example: Netflix stores popular movies on CDN edge servers around the world. When users stream a movie, the content is delivered from the nearest server instead of a distant data center, reducing buffering. Similarly, online multiplayer games use CDNs to reduce network latency and provide smoother gameplay.

Conclusion

CDNs and edge servers improve application performance, reduce latency, enhance user experience, and ensure global availability for cloud applications.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Answer

Financial institutions handle highly sensitive customer information and must comply with regulations such as PCI-DSS, ISO 27001, GDPR, and other financial security standards. Cloud providers offer security features and compliance certifications that help organizations meet these requirements.

Cloud providers support **auditing** by maintaining detailed logs of user activities, system changes, and data access. These audit logs help identify unauthorized access and support regulatory inspections.

Logging records every action performed on cloud resources. **Monitoring** continuously checks system health, detects suspicious activities, and generates alerts for security incidents. These services help organizations respond quickly to potential threats.

Encryption protects customer information while it is stored and transmitted across networks. Strong encryption algorithms ensure that sensitive financial information cannot be accessed without authorization.

To satisfy **data sovereignty** requirements, organizations can choose cloud regions where customer data must be stored. Many cloud providers allow customers to select specific countries or regions to ensure compliance with local regulations.

Example: During an online banking transaction, customer account details are encrypted before transmission. Logging records every transaction, monitoring detects unusual login attempts, and audit reports help verify compliance during regulatory reviews.

Conclusion

Cloud security, compliance certifications, encryption, logging, and monitoring enable financial institutions to protect sensitive data while meeting legal and regulatory requirements.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Answer

Containerization packages an application together with its dependencies into a lightweight container. Unlike **Virtual Machines (VMs)**, containers share the host operating system, making them much smaller and faster.

Virtual Machines require a complete guest operating system for every instance, consuming more memory, storage, and CPU resources. Containers start within seconds, while virtual machines may take several minutes to boot.

Kubernetes is a container orchestration platform that automates deployment, scaling, monitoring, and recovery of containers. If a container fails, Kubernetes automatically restarts or replaces it. It also distributes workloads across multiple servers and scales containers automatically based on demand.

Containers provide excellent **portability** because the same container image can run consistently on private clouds, public clouds, hybrid clouds, or local development systems without modification.

Example: An online shopping application follows a microservices architecture where separate containers handle user authentication, product catalog, payment processing, order management, and notifications. Kubernetes manages these containers, ensuring high availability and automatic scaling.

Conclusion

Containers provide faster deployment, lower resource usage, better portability, and easier management than virtual machines. Kubernetes further simplifies deployment, scaling, and maintenance of cloud-native applications.

